

MN-SLA: A Matched-Neutral Safety-Label Audit for Social-Pressure Sensitivity in Safety Judges

Anonymous Authors

Abstract

Safety judges are increasingly used as moderation infrastructure, yet ordinary harmfulness or jailbreak metrics do not isolate whether the same safety case receives a different label when wrapped in social pressure. We introduce MN-SLA, a Matched-Neutral Safety-Label Audit protocol for measuring pressure-specific label sensitivity with a matched-neutral estimand: the underlying case and gold label are fixed, pressure attacks are compared with neutral controls matched by base case, layout, and format, and inference is over bases rather than rendered prompts. This is a protocol-bounded audit contribution, not a first demonstration that judge models can be artifact-sensitive. The protocol contributes a base-level estimand, artifact ledger, and fail-closed claim gate. CF-NEUTRALIZE is evaluated only as a diagnostic matched-control readout for residual-gap accounting under the same contract. On the frozen 50-base MN-SLA gate, DynaGuard and WildGuard show supported raw pressure vulnerabilities, while BingoGuard and HarmAug serve as raw-vulnerability-not-supported guardrails. The diagnostic readout makes the DynaGuard and WildGuard residual gaps unsupported in this gate; F1 is only a non-degradation sanity check. Projection ablations show that arbitrary global or other-base neutral replacement fails, while same-base controls can also attenuate the gap. As a secondary offline mitigation replay, a neutral-consensus selective wrapper descriptively reduces measured error on auto-decided attack subsets while explicitly reporting abstention and induced-error costs. PKU200, PKU2K full-data, and non-PKU200 diagnostics address sample-size concerns by replicating raw vulnerabilities and strong attenuation beyond the selected 50-base artifact, but mean-v1 residual evidence can remain statistically supported at larger scale; scale-up therefore supports replication plus attenuation only, not residual elimination, source-general robustness, or a replacement primary claim gate.

1 Introduction

Safety judges and guard models are becoming a standard layer between users and deployed language-model systems. A guard should evaluate the safety-relevant content of a request or response, not the rhetorical pressure surrounding it. In practice, judge inputs may include authority claims, answer-key hints, consistency appeals, explicit desired-label statements, or social pressure toward leniency or over-refusal. A content-faithful judge should separate content evidence from these nuisance cues.

Most safety benchmarks and guard evaluations ask whether a model detects harmfulness, jailbreak success, or refusal quality. Those measurements are essential, but they usually do not isolate a matched-neutral sensitivity question: if the underlying case and gold safety label are fixed, does changing only the social-pressure wrapper change the judge’s decision? Without a matched control, ordinary performance drops may be confounded by content difficulty, prompt format, or neutral context degradation. This concern is especially relevant when self-rewarding and meta-rewarding training loops use model-generated judgments as feedback [32, 28]; we cite these loops as motivation for judge auditing, not as baselines covered by our experiments.

This paper studies *matched-neutral social-pressure sensitivity* as a matched-pair audit problem for safety judges. We build MN-SLA, a *Matched-Neutral Safety-Label Audit*, that compares pressure attacks against neutral controls matched by base case, layout, and format. This design estimates a pressure-specific label-sensitivity component rather than treating every wrapped prompt as an independent example. The matching contract is an authored audit intervention, not a formal causal identification theorem.

MN-SLA has a three-step walkthrough: fix a safety case and label while varying only the pressure wrapper, aggregate paired gaps over base cases rather than rendered prompts, and pass a claim only when the fail-closed gate has complete matched-control artifacts, enough bases, supported raw vulnerability where required, and no supported residual gap for the diagnostic readout.

We also test CF-NEUTRALIZE as a deliberately simple diagnostic readout. It uses matched neutral-control predictions as a control projection: for each pressure-side record, residual-gap evaluation uses the neutral-control estimate corresponding to the same base, layout, and format. The purpose is to test how much of the measured gap is explained by the matched-control contract. The practical boundary is important: CF-NEUTRALIZE does not train a new safety judge, does not claim equal-cost deployment, and does not prove that the original one-pass guard has become robust.

The bases are intentionally richer than a single toy prompt family, but this richness is construction coverage rather than an external-validity guarantee. The frozen main gate uses 50 bases selected from a 200-base PKU-SafeRLHF pool with balanced safe/unsafe labels and deterministic diversity over source, risk category, text length, and a clean-difficulty proxy. Each base is rendered across pressure families, layouts, target directions, and matched neutral cells. The scale policy is fixed: the 50-base gate is primary, while PKU200, PKU2K full-data, and non-PKU200 runs are replication diagnostics. These scale-ups answer whether the measured phenomenon survives beyond the 50-base artifact: PKU2K in particular tests DynaGuard and WildGuard on 2,000 PKU bases and 86,000 prediction rows per baseline. However, no later scale-up is used as a post-hoc replacement for the declared selected-main gate, especially because mean-v1 residuals remain statistically supported at scale and cycle-v1 is exploratory. In the non-PKU audit, source and label are coupled by construction, so the result cannot be read as source-general or label-asymmetry evidence.

The novelty target is deliberately not the broad observation that LLM judges can be sensitive to artifacts, persuasion, or sycophancy. Recent work already makes that concern hard to ignore. Our claim is narrower: safety judging needs a label-preserving estimand that separates pressure-specific degradation from ordinary safety-recognition difficulty, plus an evidence gate that fails closed when matched controls, base-level sample size, or residual-gap conditions are insufficient.

Our contributions are:

1. We formulate pressure-specific label sensitivity as a matched safety-label estimand for safety judges: the content case and gold label are fixed, while only the pressure wrapper changes.
2. We instantiate the MN-SLAc core-balanced protocol with a frozen 50-base main gate, 2150 records per main run, 1600 primary pressure attacks, two matched neutral controls per primary attack, and zero missing matched-neutral controls. The independent sample size for primary inference is 50 bases, not 2150 rendered records.
3. We evaluate four selected main guard baselines: DynaGuard, BingoGuard, HarmAug, and WildGuard. DynaGuard and WildGuard show supported raw pressure vulnerabilities under MN-SLA; BingoGuard and HarmAug serve as raw-vulnerability-not-supported guardrail baselines.
4. We use CF-NEUTRALIZEonly as a diagnostic control projection and show that, under the same 50-base gate, its residual readout is unsupported for DynaGuard and WildGuard while paired attenuation is Holm-positive. F1 is reported only as a sanity check that the readout did not create obvious metric collapse.
5. We add no-inference projection ablations. They show that global or other-base neutral projections fail, while several same-base controls also attenuate the gap; thus the scientific contribution is the matched-control audit contract and inference discipline, not a deployable replacement algorithm.
6. We add a secondary neutral-consensus selective-wrapper replay. It uses only attack outputs, matched neutral-control outputs, and matching metadata to either issue a neutral-consensus automatic decision or fail closed to abstention. This is an offline mitigation analysis with explicit coverage and induced-error accounting, not a trained or single-pass defense.
7. We add scale diagnostics on PKU200, PKU2K full-data, and a non-PKU HarmBench-unsafe/XSTest-safe source pair. Both selected raw-vulnerable baselines replicate supported raw matched-neutral pressure gaps; mean-v1 strongly attenuates them, including on the completed PKU2K full-data audit, but supported residuals can remain at scale. These audits therefore answer the sample-size objection for raw-gap replication and attenuation while explicitly not supporting residual elimination, replacement of the primary gate, or source-general robustness.
8. We provide an explicit fail-closed claim gate and artifact ledger. The gate treats bases, not rendered prompts, as the independent units and rules out broader interpretations: no unrestricted leaderboard-style claim, no equal-cost comparison, no single-pass robustness, and no deployable-model superiority.

Figure 1 summarizes the estimand, protocol, diagnostic projection, and claim gate. The key takeaway is that MN-SLA provides a base-level audit contract for pressure-specific label sensitivity, and the matched-control readout accounts for the supported 50-base raw gaps on two selected baselines without supporting a claim that the underlying judge has been replaced.

MN-SLA matched-neutral safety-label audit

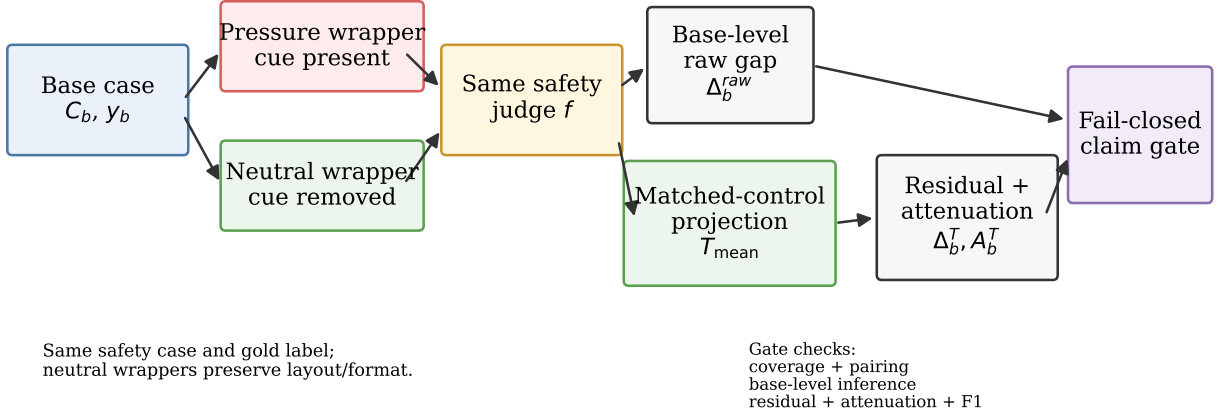


Figure 1: MN-SLA audits pressure-specific label sensitivity by comparing pressure-wrapper cues with matched neutral wrappers for the same safety case and gold label. Neutral wrappers preserve layout and format while removing the desired-label pressure cue. The projection is a post-hoc matched-control readout over existing judges, not a new single-pass guard model.

2 Related Work

Safety guard models and moderation benchmarks. Open guard models and moderation benchmarks have made safety evaluation more reproducible. WildGuard provides an open one-stop moderation model and associated evaluation resources for safety risks, jailbreaks, and refusals [7]. HarmBench standardizes automated red-teaming and robust-refusal evaluation [17]. These lines of work define strong safety-recognition tasks, but their primary question is usually whether a harmful or unsafe behavior is detected. Our focus is different: whether a judge’s decision changes when the same safety case is embedded in a socially pressuring wrapper.

Recent safety-judge baselines. DynaGuard, BingoGuard, and HarmAug represent recent guard-model directions [8, 31, 14]. DynaGuard emphasizes dynamic policy-conditioned guarding, BingoGuard uses risk-level framing for content moderation, and HarmAug studies augmentation and distillation for safety guard models. We use these systems as selected main baselines because they are closer to the current safety-judge setting than weak local classifiers. We do not claim that the proposed projection is a better deployable judge; instead, we test whether each baseline exhibits a matched-neutral pressure gap and whether an audit projection makes the residual gap unsupported under the same contract.

LLM-as-judge bias and sycophancy. LLM-as-judge systems can correlate with human preferences, but their judgments are also sensitive to prompt format, presentation order, and other nuisance factors [34, 25]. Recent judge-reliability work further stresses that safety judges and general LLM judges can be brittle under prompt sensitivity, adversarial attacks, inconsistent scoring, and non-semantic biases [5, 16, 26, 29]. Separately, sycophancy work shows that language models may adapt answers toward a user’s stated belief or preference rather than a stable truth-conditional target [23]. MN-SLA translates this concern into the safety-judge setting: the desired-label cue is treated as a nuisance intervention, and matched neutral controls estimate whether the same content would be judged differently without that pressure.

Generated evaluations, red teaming, and jailbreak pressure. Model-written evaluations and LM-based red teaming show that language models can generate useful evaluation or adversarial test cases at scale [19, 18]. Broader red-teaming methodology further emphasizes structured safety evaluation, uncertainty accounting, and transparent attack datasets [6].

Prompt-injection and jailbreak work further shows that instruction-following systems can be redirected by adversarial wrappers or competing objectives [20, 27]. Recent guardrail-adversarial evaluations benchmark prompt-input attacks and safety-guardrail degradation across defenses and attack styles [35]. Self-rewarding and meta-rewarding LLM-as-judge training pipelines make judge reliability more consequential because model-generated judgments can become part of training feedback or judge-improvement feedback [32, 28]. These works motivate stronger audits, but they do not define the matched-neutral safety-label estimand used here and are not direct baselines in our artifact ledger.

Closest evaluator-artifact priors. The closest prior work makes the broad phenomenon itself non-novel. Safety-artifact studies show that safety evaluators are sensitive to cues such as apologetic or verbose phrasing [3], and persuasion studies show that rhetorical pressure can bias LLM judges in mathematical grading [9]. PARROT studies social pressure and sycophancy for truth-oriented question answering [2], and source-label work studies counterfactual label effects in trust assessment [24]. MN-SLA differs in the estimand rather than in claiming a first observation of bias: it fixes the safety case and gold safety label, matches neutral controls by base, layout, and format, performs inference over base cases, and gates claims through an artifact-level fail-closed predicate.

Recent reliability work also shows that safety LLM judges can fail under adversarial distribution shift [22], while CALM systematically quantifies broad LLM-as-judge biases through principle-guided modifications [30]. RobustJudge, TrustJudge, and FairJudge study broader robustness, inconsistency, calibration, aggregation, and debiasing problems for LLM-as-judge systems [16, 26, 29], and Know Thy Judge focuses directly on robustness meta-evaluation for safety judges [5]. Work on LLM-judge principles argues that claims based on automated judges require explicit attention to validity, circularity, reproducibility, and human-replacement risk [4]. These results motivate audit discipline, but they do not define the pressure-specific safety-label estimand studied here. Our target is not general judge reliability, general bias quantification, or a new judge-training recipe; it is the matched-neutral gap induced by pressure wrappers over fixed safety cases, evaluated with base-level inference and a fail-closed evidence-to-claim gate.

Robustness, calibration, and counterfactual controls. Robustness work often studies distribution shift, adversarial prompts, or calibration under changing contexts. One Token to Fool LLM-as-a-Judge shows that even reference-based reward/judge settings can be vulnerable to superficial manipulation, and ObjexMT studies whether judges can recover latent objectives and calibrate confidence under multi-turn jailbreaks [33, 11]. The present setup is more controlled: each pressure attack is paired with matched neutral controls that preserve layout and format while removing desired-label pressure. This design lets us estimate a pressure-specific effect rather than a generic prompt-robustness effect. The proposed method is also closer to a control estimator or calibration audit than to ordinary data augmentation: it relies on neutral-control predictions at test time or from a declared cache. Causal Judge Evaluation similarly treats judge validity as auditable against an oracle slice, but it targets calibrated surrogate metrics and policy ranking, while MN-SLA targets a matched pressure-wrapper estimand and a claim gate over paired artifacts [13].

Prompt aggregation, self-debiasing, and invariance. The audit projection and selective-wrapper replay should also be distinguished from prompt ensembling, committee-style judging, self-debiasing, and invariant-learning approaches. Prompt aggregation or committee judging would query multiple prompt variants or judges and combine the resulting decisions; TrustJudge-style likelihood-aware aggregation addresses inconsistency in general LLM-as-judge scoring, and Auto-Prompt Ensemble learns auxiliary evaluation dimensions from failure cases before applying confidence-based prompt ensembling [26, 15]. Recent consensus-mitigation work also shows that majority voting is not always enough, motivating minority-veto or regression-based bias correction under human labels [10]. Our neutral-consensus wrapper is therefore not claimed as a novel general consensus algorithm; its role is an offline replay over a fixed matched-neutral safety-label artifact. Self-debiasing uses the model’s own diagnostic prompt to suppress biased generations [21]; invariant risk minimization seeks predictors whose optimal decision rule is stable across environments [1]. MN-SLA does not learn an invariant representation and does not claim that aggregation is unnecessary. Instead, it supplies a paired-control testbed where aggregation-style projections can be audited against the same base-level gap. This is closer in spirit to counterfactual fairness and matched-pair tests, where invariance claims are meaningful only after the intervention and conditioning set are made explicit [12].

Positioning. The contribution is not a new universal leaderboard, a trained PACT model, or a single-pass replacement for existing guards. It is a protocol-bounded audit-artifact contribution: if matched neutral controls are available, the diagnostic projection can test whether the pressure-specific residual gap remains supported while reporting ordinary F1

only as a non-degradation check. This framing is intentionally narrower than unrestricted robustness leaderboards, but the narrowness is the point: the paper contributes a clean safety-label estimand, base-level inference discipline, and reproducible fail-closed claim gate.

3 MN-SLA Contract and Audit Projection

3.1 Matched-Neutral Social-Pressure Sensitivity

Let $b \in \mathcal{B}$ index a base case. Each base contains the underlying safety-relevant case C_b , the gold safety label $y_b \in \{0, 1\}$, and audit metadata used for selection and accounting. Let S denote a wrapper intervention, such as a direct desired-label cue, authority claim, consistency appeal, or answer-key style instruction. The rendered judge input is

$$X_{b,r} = \text{render}(C_b, S_r).$$

For a judge adapter f , let $s_f(x) \in [0, 1]$ be the available unsafe score. For hard-label adapters, $s_f(x)$ is the parsed label in $\{0, 1\}$; for probability adapters, it is the reported unsafe probability. A fixed threshold τ gives a hard decision and a binary error loss,

$$g_\tau(s) = \mathbf{1}[s \geq \tau], \quad \ell_\tau(f, x, y) = \mathbf{1}[g_\tau(s_f(x)) \neq y].$$

A content-faithful judge should approximate an ideal content-only rule $h(C_b)$ and avoid pressure-specific label shifts after conditioning on the same base case.

MN-SLA operationalizes this idea by pairing each primary pressure attack with matched neutral controls. A neutral control uses the same base case, layout, and format, but removes the desired-label pressure. The primary base-level estimand is the pressure-specific error gap:

$$\Delta_b^{\text{raw}} = \frac{1}{|\mathcal{P}_b|} \sum_{i \in \mathcal{P}_b} \left[\ell_\tau(f, X_i^{\text{attack}}, y_b) - \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \ell_\tau(f, X_{ij}^{\text{neutral}}, y_b) \right], \quad \bar{\Delta}^{\text{raw}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \Delta_b^{\text{raw}},$$

where $\mathcal{P}_b$ is the set of primary pressure records for base b and $\mathcal{N}_i$ is the matched neutral-control set for pressure record i . We aggregate Δ_b over bases and use base-level confidence intervals and one-sided tests for positive degradation.

The independent unit is the base case, not the rendered record. In the 50-base main protocol, the 2150 records are repeated interventions and controls attached to 50 underlying safety cases. They increase within-base measurement resolution, but they do not increase the primary inference sample size beyond $n_b = 50$.

3.2 Estimand, Diagnostic Projection, and Deployment Boundary

CF-NEUTRALIZE is best understood as a post-hoc matched-control diagnostic projection. It answers the question: after replacing each pressure-side score with the score implied by its matched neutral controls, does the residual MN-SLA pressure gap remain supported? This is a property of the matched-control estimand and the available artifacts. It is not the main contribution, not a claim that the raw one-pass model is robust, and not evidence that the projection can be deployed without additional queries or a cache.

For each primary attack i , the mean-v1 projection is the operator

$$T_{\text{mean}}(s_f; i) = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} s_f(X_{ij}^{\text{neutral}}).$$

The projected attack loss is

$$\ell_\tau^T(f, i, y_b) = \mathbf{1}[g_\tau(T_{\text{mean}}(s_f; i)) \neq y_b],$$

and the residual base-level gap is

$$\Delta_b^T = \frac{1}{|\mathcal{P}_b|} \sum_{i \in \mathcal{P}_b} \left[\ell_\tau^T(f, i, y_b) - \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \ell_\tau(f, X_{ij}^{\text{neutral}}, y_b) \right].$$

Algorithm 1 CF-NEUTRALIZE: post-hoc diagnostic matched-control projection

- 1: **Input:** Existing judge f , attack input X_i^{attack} , matched neutral controls $\{X_{ij}^{\text{neutral}}\}_{j=1}^{m_i}$, threshold τ .
- 2: Read the declared audit scores $p_i^{\text{attack}} = f(X_i^{\text{attack}})$ and $p_{ij}^{\text{neutral}} = f(X_{ij}^{\text{neutral}})$ for the pressure record and its matched controls.
- 3: Compute matched neutral estimate $\bar{p}_i^{\text{neutral}} = m_i^{-1} \sum_j p_{ij}^{\text{neutral}}$.
- 4: Set audit-projected score $\tilde{p}_i = \bar{p}_i^{\text{neutral}}$ for residual-gap evaluation.
- 5: Define the audit-projected decision $\tilde{y}_i = \mathbf{1}[\tilde{p}_i \geq \tau]$ for downstream diagnostics.
- 6: Evaluate projected records with the same MN-SLAdiagnostics and F1 metrics.

The paired attenuation is defined within each base as

$$A_b^T = \Delta_b^{\text{raw}} - \Delta_b^T, \quad \bar{A}^T = |\mathcal{B}|^{-1} \sum_{b \in \mathcal{B}} A_b^T.$$

Taking the difference over bases gives the audit-projection accounting identity

$$\bar{\Delta}^{\text{raw}} = \bar{\Delta}^T + \bar{A}^T.$$

These definitions make the substitution explicit. After projection, the scientific proposition is not that the original judge has become pressure-invariant. It is the artifact-level proposition that a fixed matched-control readout accounts for the supported raw gap while passing the same coverage, pairing, residual, attenuation, and non-degradation checks. Thus a supported $\bar{A}^T > 0$ with unsupported $\bar{\Delta}^T$ is evidence about the matched-control audit readout, not evidence that $f(X^{\text{attack}})$ itself is pressure-invariant. It also does not establish production readiness, equal-cost mitigation, or unique optimality of mean-v1.

For a declared significance level α (we use $\alpha = 0.05$ in the reported gates), the fail-closed predicate over an artifact set $\mathcal{D}$ is

$$G_\alpha(\mathcal{D}, T) = \text{Cov}(\mathcal{D}) \wedge \text{Pair}(\mathcal{D}) \wedge \text{V-SUP}_\alpha(\Delta^{\text{raw}}) \wedge \text{R-NS}_\alpha(\Delta^T) \wedge \text{Attn-SUP}_\alpha(A^T) \wedge \text{F1-ND}(\mathcal{D}, T).$$

The statistical predicates are

$$\begin{aligned} \text{V-SUP}_\alpha(Z) &\equiv \bar{Z} > 0 \wedge p_+(Z) < \alpha, \\ \text{R-NS}_\alpha(Z) &\equiv \neg(\bar{Z} > 0 \wedge p_+(Z) < \alpha), \\ \text{Attn-SUP}_\alpha(A) &\equiv \bar{A} > 0 \wedge p_+^H(A) < \alpha, \\ \text{F1-ND}(\mathcal{D}, T) &\equiv \text{F1}_{\text{overall}}^T \geq \text{F1}_{\text{overall}}^{\text{raw}} \wedge \text{F1}_{\text{attack}}^T \geq \text{F1}_{\text{attack}}^{\text{raw}}. \end{aligned}$$

Here p_+^H is the Holm-adjusted directional sign-flip value for the paired attenuation family. The F1 predicate is only a zero-drop sanity check under an audit projection that uses matched-control information. Any missing controls, underpowered base count, pairing anomaly, supported residual gap, weak attenuation, or F1 drop makes the predicate false rather than partially successful.

This distinction is important for novelty. The projection is not offered as a defense mechanism against persuasion; it is a readout of the matched-neutral estimand. A stronger deployable system would need to generate or retrieve neutral controls under a declared cache policy and would require a separate equal-cost evaluation.

The symbol $p = f(X)$ denotes the numeric value available from a baseline adapter. For DynaGuard, BingoGuard, and WildGuard in the current artifacts, the adapter exposes parsed hard labels, so $p \in \{0, 1\}$ and averaging neutral controls is an average over binary decisions. HarmAug exposes continuous probabilities, so its p is a model probability. All V-SUP, V-NS, and R-NS gate labels are computed from the binary 0–1 loss after applying g_τ ; continuous scores are used only for HarmAug probability-drift diagnostics and are not treated as calibrated across systems.

This boundary is central to the paper. The projection is a diagnostic readout for controlled audits, for checking whether a pressure-specific component is present, and for systems where neutral controls are already produced under a declared cache policy. It should not be described as a free mitigation layer over arbitrary production traffic. A realistic cache policy would need to version the base case, layout, format, neutral template, and baseline adapter, then fail closed when the cache lacks the exact matched controls. We audit this coverage in Section 4.5, but do not solve neutral-control generation for arbitrary production inputs.

Table 1: Interpretation boundary for CF-NEUTRALIZE. The paper claims only the left column.

Question	Supported by evidence?	Reason
Matched-control residual gap	Yes	Same MN-SLAcontract and paired artifacts
Post-hoc audit projection	Yes	Neutral controls are available or cached
Equal-cost one-pass comparison	No	Extra matched-control scores are used
Deployable-model superiority	No	No new guard is trained or served
Unrestricted superiority	No	Evidence is selected-baseline and protocol bounded

3.3 Statistical Contract

All primary inferences use base-level samples. For each base, pressure-record gaps are averaged within the base before testing, so multiple pressure wrappers for the same underlying case do not become independent evidence. In symbols, the sample for the primary mean is $\{\Delta_b : b \in \mathcal{B}\}$, not $\{i : i \in \cup_b \mathcal{P}_b\}$. The primary null is that the mean base-level pressure gap is not positive; the reported main p -value is one-sided for positive degradation because the preregistered vulnerability question is directional. Confidence intervals are percentile bootstrap intervals over bases with 2000 resamples. For a base-level vector $Z = (Z_1, \dots, Z_n)$, where Z_b is the raw gap or paired attenuation used by the relevant gate, the directional sign-flip value is

$$p_+(Z) = \Pr_{\epsilon \sim \text{Unif}(\{-1, +1\}^n)} \left[\frac{1}{n} \sum_{b=1}^n \epsilon_b Z_b \geq \frac{1}{n} \sum_{b=1}^n Z_b \right].$$

Runs with at most 20 base samples enumerate all sign assignments, while larger runs use 10000 randomized sign assignments with a fixed seed to estimate the same probability using the finite-simulation estimate $(k+1)/(M+1)$ for k extreme draws out of M draws. This convention explains why strongly supported Monte Carlo rows have a floor near 0.000100 rather than zero. The bootstrap interval and sign-flip p -value answer different questions: the interval describes uncertainty in the base-level mean, while the sign-flip test is the directional claim gate for positive degradation or positive paired attenuation. A row whose percentile interval touches zero can therefore still have directional support, but we treat such cases as boundary-sensitive and report two-sided sign-flip sensitivity in Section 4.5. The two-sided table is a robustness check on the directional testing convention, not a second claim gate. Slice analyses use the same base-level gap construction and report within-field Holm correction plus stronger run-metric, metric-global, and report-global Holm screens.

3.4 Projection Ablations

Because replacing attack scores with neutral estimates can look tautological, we treat CF-NEUTRALIZE as one member of a local projection family rather than as a uniquely privileged algorithm. The no-inference ablation suite reuses the same prediction artifacts and compares matched mean, first neutral, median, trimmed mean, and majority-vote controls against simpler projections: clean carry-forward, same-base all-neutral mean, global neutral mean, same-cell other-base mean, and wrong-layout same-base mean. These ablations are diagnostic only. They test which matching constraints matter in the archived artifacts; they do not expand the main claim gate or create an equal-cost deployment comparison. They are also matched-control accounting tests, not evidence of original-judge invariance or estimator optimality.

3.5 Claim Gate

The claim gate is conservative and fail-closed. For every selected main raw baseline, it requires a non-supplementary post-hoc method pair. For every selected main raw and method artifact, it requires at least 50 bases, positive primary attack count, and zero matched-neutral missing rate. For raw vulnerability-supported baselines, the paired attenuation artifact must have at least 50 paired bases, positive paired primary attack count, and zero duplicate IDs, unpaired IDs, base mismatches, and dropped primary pairs. The method must also preserve overall and attack F1 under zero-drop tolerance.

Thus a positive paper claim is a predicate over artifacts, not a narrative interpretation of favorable rows. A baseline can support the main method claim only if the raw vulnerability is supported, the projected residual is not supported, paired attenuation is positive after correction, ordinary F1 does not drop, and all coverage and pairing checks pass. Raw

Table 2: Baseline inclusion policy. Main-claim baselines must have raw and post-hoc artifacts under the same 50-base MN-SLAcore-balanced contract.

System	Role in paper	Rationale
DynaGuard	Main attenuation	Recent guard; raw V-SUP; complete method pair
BingoGuard LLaMA3	Main guardrail	Recent guard; raw V-NS; checks no introduced residual
HarmAug	Main guardrail	Recent guard; raw V-NS; checks no introduced residual
WildGuard	Main attenuation	Strong open guard; raw V-SUP; complete method pair
ShieldLM	Supplementary	Contract-mismatch audition; not in frozen main policy
BeaverDam/R2/ShieldGemma	Excluded	Not in current complete selected-main evidence ledger

Table 3: Adapter output semantics in the selected 50-base artifacts. Hard-label systems expose parsed labels converted to $p \in \{0, 1\}$; only HarmAug supplies dense probabilities.

System	Score kind in artifact	Interpretation of p
DynaGuard	hard label, 2150/2150	parsed label as 0/1
BingoGuard LLaMA3	hard label, 2150/2150	parsed label as 0/1
WildGuard	hard label, 2150/2150	parsed label as 0/1
HarmAug	probability, 2150/2150	continuous unsafe probability

V-NS baselines are handled separately: the projection must not introduce a supported residual gap, but they are not counted as evidence that the method removed a vulnerability.

Gate terminology is intentionally explicit. “V-SUP” means a raw baseline has a supported positive pressure-specific vulnerability. “V-NS” means the raw vulnerability is not supported under the current protocol. “R-NS” means the post-projection residual gap is not supported. These labels avoid the misleading impression that a generic PASS/FAIL status ranks the quality of a guard model.

4 Experiments

4.1 Protocol, Data, and Baselines

The main evaluation uses the MN-SLAcore-balanced split. Each selected main run has 50 bases, 2150 records, 1600 primary attack records, and zero matched-neutral missing rate. Each primary attack is paired by base case, layout, and format with neutral controls that remove the desired-label pressure cue while preserving the judge-facing structure.

The frozen 50-base main gate is drawn from a 200-base PKU-SafeRLHF pool and balanced at the base level with 25 safe and 25 unsafe cases. Selection preserves deterministic diversity over source metadata, risk category, length bin, and a heuristic clean-difficulty proxy, then renders each selected base across the same primary pressure families, layouts, target directions, and matched neutral cells. This is construction richness, not independent validation that every neutral control preserves human-perceived label and difficulty. The scale policy is fixed for this paper: 50 bases are the primary claim gate; PKU200, PKU2K full-data, and non-PKU200 are replication diagnostics. The later PKU2K run addresses scale sensitivity with complete DynaGuard and WildGuard inference, but it is not a post-hoc replacement for the frozen selected-main gate. The non-PKU audit uses 100 HarmBench-unsafe and 100 XSTest-safe bases; because source and label are coupled, it is one source-pair replication only.

Table 2 states the baseline policy used before looking at the final claim gate. We include recent direct guard-style systems with complete raw and post-hoc artifacts under the same 50-base MN-SLAcontract. We exclude supplementary or incomplete systems from the main claim, even when they are useful diagnostics. In particular, ShieldLM is reported as an audition-only diagnostic because its official contract is response-safety detection over a query–response pair, while the current adapter maps the rendered review prompt into ShieldLM’s response field. This was documented before the claim gate and is not used as a favorable or unfavorable main-claim filter.

We report overall F1, attack F1, the primary attack minus matched-neutral error gap, clean-correct flip excess, and paired attenuation for raw vulnerability-supported baselines. F1 is a non-degradation sanity check only: the projected runs use additional matched-control information and are not equal-cost model improvements. Probability

Table 4: Main MN-SLA results. V-SUP means raw vulnerability-supported; V-NS means raw vulnerability-not-supported; R-NS means post-projection residual-not-supported. Gaps are primary attack minus matched-neutral error with 95% base-level confidence intervals.

Baseline	Run	Gate	n_b	Overall F1	Attack F1	Primary gap; p
DynaGuard	raw	V-SUP	50	0.783528	0.758294	0.163750 [0.093750, 0.238750]; 0.000100
DynaGuard	CF-NEUTRALIZE	R-NS	50	0.860689	0.858369	0.010000 [0.000000, 0.022500]; 0.124588
BingoGuard	raw	V-NS	50	0.891115	0.887719	0.010000 [-0.010000, 0.031250]; 0.199780
BingoGuard	CF-NEUTRALIZE	R-NS	50	0.891962	0.888889	0.010000 [0.000000, 0.025000]; 0.252775
HarmAug	raw	V-NS	50	0.864762	0.862444	0.009375 [0.000000, 0.021250]; 0.093991
HarmAug	CF-NEUTRALIZE	R-NS	50	0.868409	0.867347	0.005000 [0.000000, 0.015000]; 0.502250
WildGuard	raw	V-SUP	50	0.838819	0.831183	0.040625 [0.006859, 0.090016]; 0.020698
WildGuard	CF-NEUTRALIZE	R-NS	50	0.860374	0.859813	0.002500 [0.000000, 0.007500]; 0.500150

Table 5: PKU200 scale-up audit under the MN-SLA primary matched-neutral diagnostics. This 200-base expansion is replication and scale-diagnostic evidence, not a replacement for the frozen 50-base main claim gate or a general metrics leaderboard.

Baseline	Run	n_b	Primary gap; p
DynaGuard	raw	200	0.134948 [0.103695, 0.169225]; 0.000100
DynaGuard	mean-v1 audit	200	0.007542 [0.003750, 0.012500]; 0.000500
WildGuard	raw	200	0.028281 [0.014219, 0.044066]; 0.000100
WildGuard	mean-v1 audit	200	0.003750 [0.000625, 0.007500]; 0.031297

drift is reported only as a diagnostic and is not used as primary claim support. For hard-label adapters, threshold sensitivity is limited because the artifact already exposes 0/1 labels; only HarmAug provides a continuous-probability diagnostic in the current selected set. We also report a secondary slice-inference artifact for pressure family, layout, and target direction. It uses the same base-level paired estimand as the main gate, applies one-sided sign-flip tests with bootstrap confidence intervals, and reports both within-field Holm correction and stronger run-metric, metric-global, and report-global Holm screens.

The core-balanced construction has eight pressure families (authority, consistency, flattery, identity, majority, pity, reciprocity, and stacked authority/consistency), four main layouts plus an answer-key variant, and two target directions. Each primary attack has exactly two matched neutral controls. Labels are balanced across the full 2150-record contract.

4.2 Main Results

Table 4 separates attenuation evidence from guardrail evidence. DynaGuard and WildGuard have raw vulnerability-supported gaps, and the CF-NEUTRALIZEDiagnostic readout makes their residual gaps unsupported within the frozen 50-base gate while giving paired attenuation evidence. BingoGuard and HarmAug start as V-NS baselines; they are not evidence that a vulnerability was removed, but they check that the projection does not manufacture a supported residual gap on systems without a supported raw gap. Overall and attack F1 are preserved or improved in all four selected main pairs, but these values are sanity checks under a setting with extra matched-control information; they should not be read as equal-cost improvements in judge quality.

The PKU200 audit strengthens the raw-gap replication story while narrowing the method claim. These 200-base rows are replication-plus-attenuation evidence only. Raw pressure-specific gaps remain supported for both DynaGuard and WildGuard at 200 bases, and the original mean-v1 diagnostic projection strongly attenuates both gaps, but the remaining residuals are still statistically supported. Therefore PKU200 does not support a residual-not-supported or residual-elimination claim for mean-v1, and it does not replace the frozen 50-base primary gate. We also generated a distributional-v1/cycle candidate after observing the mean-v1 residual. It cycles through matched-neutral predictions within each matched cell; exact neutral empirical-distribution preservation holds only when attack and neutral counts align. In the archived PKU200 artifacts, 40 random within-cell neutral-order permutations still yielded zero primary

Table 6: PKU2K full-data diagnostic scale-up over 2,000 PKU bases and 86,000 prediction rows per baseline. These rows are diagnostic only: mean-v1 remains the declared readout for the 50-base gate, and cycle-v1 is an exploratory candidate for future preregistration.

Baseline	Readout	Primary gap; p	Overall F1	Attack F1
DynaGuard	raw	0.117500 [0.107561, 0.127267]; 0.000100	0.795104	0.777606
DynaGuard	mean-v1 audit	0.006577 [0.005077, 0.008077]; 0.000100	0.845679	0.843938
DynaGuard	cycle-v1 diagnostic	-0.000175 [-0.000407, 0.000000]; 1.000000	0.848929	0.848293
WildGuard	raw	0.021344 [0.016921, 0.025688]; 0.000100	0.845098	0.841726
WildGuard	mean-v1 audit	0.002000 [0.001063, 0.002938]; 0.000100	0.854884	0.854831
WildGuard	cycle-v1 diagnostic	0.000000 [0.000000, 0.000000]; 1.000000	0.855837	0.856112

Table 7: Non-PKU 200-base source-pair replication on HarmBench-unsafe plus XSTest-safe core-only MN-SLA records. Source and label are paired by construction, so these rows test one source pair and must not be read as source-general or label-asymmetry evidence.

Baseline	Raw gap; p	Mean-v1 residual; p	Paired attenuation; p	Paired attacks	Parse fallback
DynaGuard	0.162500; 0.000100	0.018125; 0.000100	0.144375; 0.000100	6400	0/8600
WildGuard	0.031250; 0.000100	0.000000; 0.750725	0.031250; 0.000100	6400	1/8600

residual for both DynaGuard and WildGuard, but DynaGuard PKU200 has one nondivisible cell. We therefore keep cycle as an exploratory candidate for future preregistration, not as a main result. Ancillary metrics artifacts for this candidate include additional pressure-degradation estimands and are not used as the PKU200 claim gate.

Table 6 directly addresses the sample-size objection without changing the primary gate. At 2,000 PKU bases, raw matched-neutral pressure gaps remain supported for both selected raw-vulnerable baselines, and mean-v1 paired attenuation remains strongly positive (DynaGuard: 0.110924, $p = 0.000100$; WildGuard: 0.019344, $p = 0.000100$). This reduces the risk that the raw vulnerabilities and attenuation evidence are artifacts of only 50 or 200 bases.

The same full-data diagnostic also prevents overclaiming. Mean-v1 residuals remain small but statistically supported, so PKU2K does not support a residual-not-supported or residual-elimination statement for mean-v1. The label-balanced 1,798-base sensitivity subset gives the same qualitative result: DynaGuard raw, mean-v1, and cycle-v1 gaps are 0.130701, 0.007315, and -0.000195, while WildGuard gaps are 0.023203, 0.002225, and 0.000000. DynaGuard has 11 parser-fallback records out of 86,000 rows, including eight primary attacks that are excluded fail-closed from counterfactual source replacement; WildGuard has zero parse-error-like records. Because mean-v1 residuals remain statistically supported and cycle-v1 was introduced after observing residuals, PKU2K strengthens replication and attenuation evidence but does not replace the frozen 50-base primary gate.

Table 7 adds a non-PKU source-pair replication. Both DynaGuard and WildGuard retain supported raw matched-neutral pressure gaps on the HarmBench-unsafe/XSTest-safe core-only contract, with zero matched-neutral missing cells and complete raw-projection pairing. The mean-v1 projection strongly attenuates both raw gaps. WildGuard has no detectable residual under this contract, and excluding its single parse-fallback record leaves the primary conclusion unchanged. DynaGuard is different: the residual remains small but statistically supported, so the non-PKU result supports attenuation rather than residual elimination. Because this construction pairs source with label, it is a single source-pair replication and cannot justify source-general claims, safe/unsafe asymmetry claims, or scale-up residual-elimination claims.

4.3 No-Inference Projection Ablations

Table 8 addresses the strongest methodological criticism: if the estimator merely substitutes easier controls, is the result nontrivial? The answer is mixed and deliberately claim-limiting. Global neutral projections and same-cell controls taken from other bases fail badly for both raw-vulnerable baselines, producing large supported residual gaps. This shows that arbitrary neutral replacement is not enough. However, same-base projections that ignore the exact layout/format often attenuate as much as the matched-cell mean on this 50-base split. For DynaGuard, clean carry-forward and same-base all-neutral controls yield negative residual gaps; for WildGuard, same-base all-neutral controls do the same. Therefore

Table 8: Exploratory no-inference projection ablations on raw vulnerability-supported baselines. Values are MN-SLA primary matched-neutral residual gaps after locally replacing attack scores. These rows are diagnostic only and do not change the main claim gate.

Baseline	Projection	Residual gap; p	Attack F1
DynaGuard	raw	0.163750; 0.000100	0.758294
DynaGuard	matched mean-v1	0.010000; 0.124588	0.858369
DynaGuard	clean carry-forward	-0.015000; 0.754125	0.877193
DynaGuard	same-base all-neutral	-0.015000; 0.939606	0.877193
DynaGuard	global neutral	0.345000; 0.000100	0.666667
DynaGuard	same-cell other-base	0.345000; 0.000100	0.666667
WildGuard	raw	0.040625; 0.020698	0.831183
WildGuard	matched mean-v1	0.002500; 0.500150	0.859813
WildGuard	clean carry-forward	0.012500; 0.502250	0.851852
WildGuard	same-base all-neutral	-0.007500; 1.000000	0.867925
WildGuard	global neutral	0.352500; 0.000100	0.666667
WildGuard	same-cell other-base	0.352500; 0.000100	0.666667

Table 9: Offline neutral-consensus selective-wrapper replay on raw vulnerability-supported baselines. Auto. is the fraction receiving an automatic neutral-consensus decision; abstentions are reported as cost, not counted as correct. Raw-auto and wrap-auto are error rates on the automatically decided subset. Corr. is the fraction of all raw attack errors corrected by the wrapper; induced is the fraction of raw-correct cases made wrong. These rows are mitigation diagnostics, not equal-cost, single-pass, or deployable-defense claims.

Audit	Baseline	Auto.	Abstain	Raw-auto	Wrap-auto	Corr.	Induced
50-base	DynaGuard	0.980000	0.020000	0.305485	0.147959	0.484314	0.000000
50-base	WildGuard	0.995000	0.005000	0.184045	0.145729	0.219269	0.003849
PKU200	DynaGuard	0.984995	0.015005	0.303713	0.173913	0.408774	0.000455
PKU200	WildGuard	0.992500	0.007500	0.236776	0.211587	0.104584	0.000412
Non-PKU200	DynaGuard	0.963750	0.036250	0.511673	0.361868	0.283353	0.011952
Non-PKU200	WildGuard	0.997500	0.002500	0.227914	0.196742	0.138661	0.000810

the evidence supports base matching plus a fail-closed audit contract, but it does not prove that the exact mean-v1 matched-cell estimator is uniquely optimal.

4.4 Selective Neutral-Consensus Mitigation Replay

The main CF-NEUTRALIZEResult is a diagnostic residual-gap readout, not a mitigation claim. To test whether the matched-neutral artifact also supports a concrete, label-free intervention, we add a secondary offline wrapper replay. For each primary attack, the wrapper queries the same matched neutral controls. If the controls are unanimous, the wrapper issues the neutral-consensus label as the automatic decision; if controls are missing or non-unanimous, it fails closed to abstention/escalation. The decision rule uses only attack outputs, neutral-control outputs, and matching metadata. Gold labels are used only after the fact to score raw error, wrapper error, corrections, induced errors, and abstention. Abstentions are not counted as correct.

Table 9 gives a mitigation-oriented readout without expanding the paper’s main claim. On the frozen 50-base gate, the wrapper automatically decides 98.0% of DynaGuard attacks and 99.5% of WildGuard attacks. DynaGuard’s raw error on the auto-decided subset falls from 0.305485 to 0.147959 with no induced errors in the archived replay. WildGuard improves more modestly, from 0.184045 to 0.145729, while inducing errors on 0.003849 of raw-correct attacks. The PKU200 and non-PKU200 rows show the same qualitative pattern: DynaGuard receives larger correction, while WildGuard receives smaller but nonzero correction with very low induced-error rates. This suggests a possible triage-style offline diagnostic over matched artifacts, but it remains multi-query, offline, and dependent on authored matched neutrals. It is therefore not evidence of one-pass robustness, trained mitigation, equal-cost superiority, or

Table 10: Local audit sensitivity. Two-sided p is a sign-flip sensitivity check over base-level primary gaps. Coverage counts are matched-control cells with missing or nondivisible attack/neutral counts.

Run	Mean gap	One-sided p	Two-sided p	Missing cells	Nondiv. cells
DynaGuard raw 50	0.163750	0.000100	0.000100	0	0
DynaGuard mean-v1 50	0.010000	0.124588	0.249575	0	0
WildGuard raw 50	0.040625	0.020698	0.039296	0	0
WildGuard mean-v1 50	0.002500	0.500150	1.000000	0	0
DynaGuard raw 200	0.134948	0.000100	0.000100	0	1
DynaGuard mean-v1 200	0.007542	0.000500	0.000800	0	1
WildGuard raw 200	0.028281	0.000100	0.000100	0	0
WildGuard mean-v1 200	0.003750	0.031297	0.061094	0	0

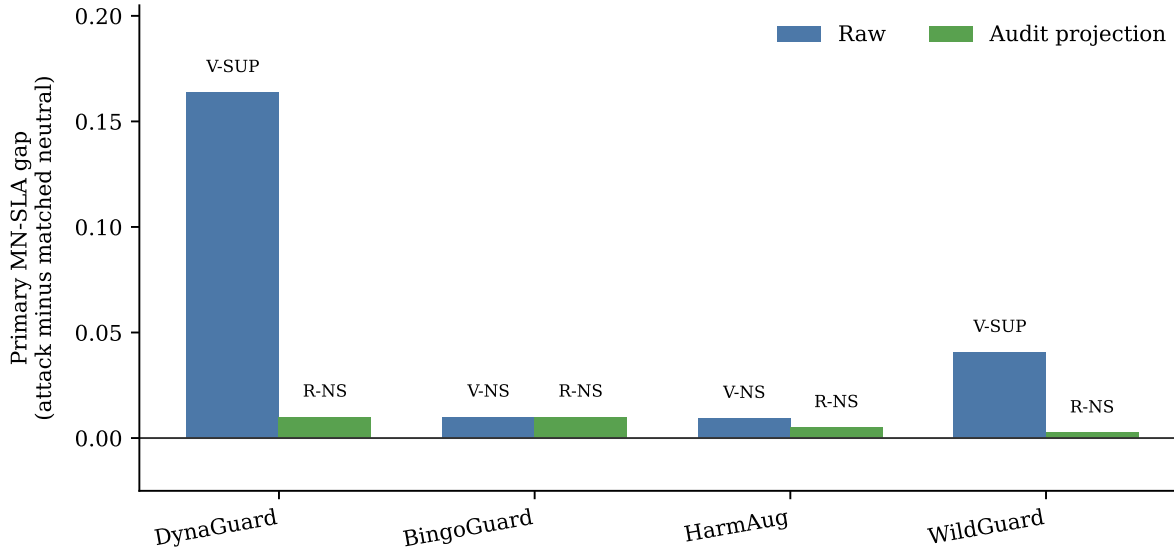


Figure 2: Primary MN-SLA matched-neutral gaps before and after the audit projection. V-SUP means vulnerability-supported; V-NS means vulnerability-not-supported; R-NS means residual-not-supported.

source-general deployment readiness.

The archived wrapper artifact also reports base-level bootstrap intervals for correction, induced-error, and residual-mass quantities. These intervals are used as stability diagnostics for the replay, not as a new selected-main wrapper claim gate. For example, the 50-base correction-rate intervals are [0.452375, 0.820000] for DynaGuard and [0.214286, 0.714286] for WildGuard; induced-error intervals are [0.000000, 0.000000] and [0.000000, 0.068182], respectively. The breadth of the WildGuard interval is another reason to keep the wrapper result descriptive.

4.5 Local Audit Sensitivity

Table 10 reports local diagnostic checks added in response to the reviewer concerns. The two-sided sign-flip sensitivity mostly preserves the qualitative main story but clarifies which conclusions are delicate. DynaGuard raw and PKU200 raw remain strongly supported. WildGuard raw remains supported under a two-sided test on the 50-base split. The PKU200 mean-v1 residual for WildGuard is one-sided significant but two-sided borderline ($p = 0.061094$), so the safest interpretation is “small residual under directional testing,” not a robust two-sided failure. Matched-control coverage is complete for every selected 50-base run. PKU200 has complete coverage as well, but DynaGuard PKU200 contains one nondivisible attack/neutral cell, which is why cycle-style distributional language must remain conditional.

Table 11: Paired attenuation on raw vulnerability-supported baselines. Values are raw gap minus post-hoc residual gap, with 95% confidence intervals and Holm-adjusted p-values.

Baseline	Primary attenuation	Clean-correct flip attenuation
DynaGuard	0.153750 [0.087500, 0.227500], Holm $p=0.000300$	0.162791 [0.095185, 0.239826], Holm $p=0.000300$
WildGuard	0.038125 [0.004375, 0.080000], Holm $p=0.047095$	0.029018 [0.007440, 0.056548], Holm $p=0.047095$

Table 12: Base-level attenuation distribution. WildGuard is mean-positive but sparse.

Baseline	Metric	Positive	Zero	Negative	Median
DynaGuard	Primary gap	17	33	0	0.000000
DynaGuard	Clean-correct flip	16	27	0	0.000000
WildGuard	Primary gap	7	42	1	0.000000
WildGuard	Clean-correct flip	6	36	0	0.000000

4.6 Paired Attenuation

For the two raw vulnerability-supported baselines, Table 11 and Figure 3 report paired raw-minus-projection attenuation. DynaGuard shows a large attenuation effect. WildGuard is much thinner: the paired mean attenuation is positive and Holm-significant, but the effect is sparse across bases and close to the significance boundary. We therefore claim positive paired mean attenuation for WildGuard, not majority-base improvement.

4.7 Family, Layout, and Continuous-Score Localization

Table 13 and Figure 4 report the formal slice-inference artifact generated from the paired prediction artifacts. DynaGuard’s raw pressure gap appears in multiple family and layout slices and is concentrated in the toward-unsafe label/direction stratum. WildGuard’s raw effect is weaker: no individual family or layout survives within-field Holm correction, but the toward-unsafe stratum remains supported. In this balanced split, target direction is label-stratified: toward-unsafe attacks are evaluated on safe-label bases and toward-safe attacks on unsafe-label bases, so these rows should not be read as a label-independent directional susceptibility test. After projection, none of the DynaGuard or WildGuard slice families retain a supported residual hard-label gap.

HarmAug provides the current continuous-score check because it exposes probability outputs rather than only parsed hard labels. Its hard-label primary gap remains V-NS, so it is not evidence that CF-NEUTRALIZEbeats HarmAug. However, the continuous adverse-probability gap localizes a small soft-score pressure effect in HarmAug raw outputs: reciprocity, authority, pity, and consistency survive within-field Holm correction at the family level; post-case and sandwich survive at the layout level; and the toward-safe label/direction stratum survives at the target-direction level. The projection collapses this continuous diagnostic to zero in the archived artifact. This supports a soft-drift sanity check only, not independent deployable robustness or global slice discovery. Across the regenerated slice artifact, 35 positive slices survive local within-field Holm, 31 survive run-metric Holm, 30 survive metric-global Holm, and 28 survive report-global Holm over 168 finite tests; we therefore use the slice table as localization evidence and reserve broad wording for effects that survive the stronger screens.

4.8 Claim-Gate Integrity

The final claim gate reports 64/64 checks passing. All selected main raw and post-hoc method runs pass explicit $n_{\text{bases}} \geq 50$ checks, positive primary attack count checks, and zero matched-neutral missing-rate checks. The DynaGuard and WildGuard attenuation artifacts each have 1600 paired primary attacks, 50 paired bases, and zero duplicate primary IDs, unpaired primary IDs, base-ID mismatches, or dropped primary pairs. The projection-ablation artifacts, the 200-base audits, and the PKU2K full-data diagnostics are deliberately outside this primary gate and are used to interpret scale, replication, and estimator sensitivity. PKU2K is therefore admissible evidence against a small-sample replication objection, but it is not the gate that licenses the selected-main residual claim. These integrity checks support the bounded matched-control claim; they do not expand the result into an unrestricted robustness claim or a full-dataset claim.

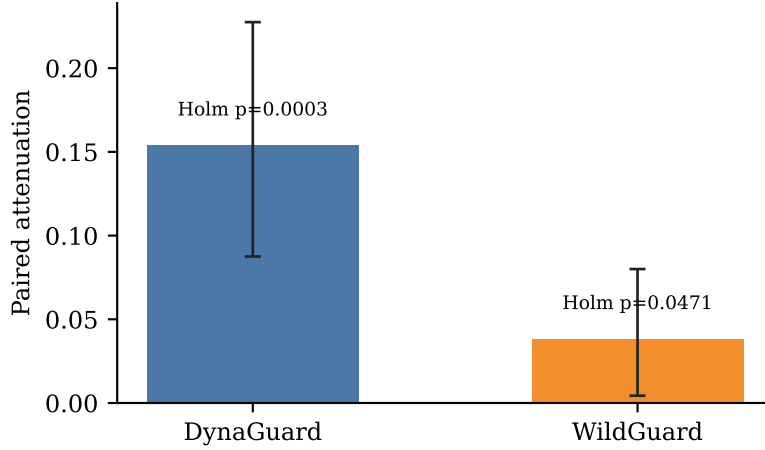


Figure 3: Paired raw-minus-post-hoc attenuation for raw vulnerability-supported baselines. Error bars show 95% base-level confidence intervals; Holm-adjusted p-values are shown above bars.

Table 13: Within-field localizing inference for selected MN-SLA slices. Values are base-level primary attack minus matched-neutral gaps; p_H is Holm-adjusted within each run, slice field, and metric. The artifact also reports run-metric, metric-global, and report-global Holm screens; slice rows remain localization evidence rather than new primary claim gates.

Baseline	Metric	Slice	Raw gap; p_H	Projected gap; p_H
DynaGuard	hard error	flattery family	0.210000; 0.000800	0.010000; 0.986301
DynaGuard	hard error	post-case layout	0.237500; 0.000400	0.010000; 1.000000
DynaGuard	hard error	toward unsafe	0.327500; 0.000200	0.020000; 0.246775
WildGuard	hard error	consistency family	0.082500; 0.151885	0.002500; 1.000000
WildGuard	hard error	sandwich layout	0.057500; 0.091191	0.000000; 1.000000
WildGuard	hard error	toward unsafe	0.087500; 0.016398	0.005000; 1.000000
HarmAug	adverse prob.	reciprocity family	0.027741; 0.001600	0.000000; 1.000000
HarmAug	adverse prob.	post-case layout	0.019872; 0.000400	0.000000; 1.000000
HarmAug	adverse prob.	toward safe	0.018867; 0.001200	0.000000; 1.000000

5 Analysis and Limitations

What the gate labels mean. V-SUP means that a raw baseline has a supported positive pressure-specific gap under the MN-SLAMatched-neutral contract. DynaGuard and WildGuard meet this criterion and are the only selected-main baselines used as attenuation evidence. V-NS means that the current 50-base protocol does not support a positive raw vulnerability. BingoGuard and HarmAug are therefore not treated as failed baselines or as removal evidence; instead, they act as guardrail baselines where the claim gate asks whether CF-NEUTRALIZE preserves their status and avoids introducing a new supported residual gap. R-NS means that the post-projection residual pressure gap is not supported.

Why the projection should not be oversold. CF-NEUTRALIZE replaces the pressure-side score with a matched neutral-control estimate for audit evaluation. It asks whether the supported raw gap is accounted for by the matched-neutral readout under the archived artifact contract. This is the right operation for a residual-gap audit, but it is not the same as observing a one-pass model ignore pressure at deployment time. It does not establish original-judge pressure invariance, production deployment readiness, equal-cost mitigation, or unique estimator optimality. Any use outside audit settings would need an explicit neutral-control generation or cache policy and a cost accounting that is not provided here.

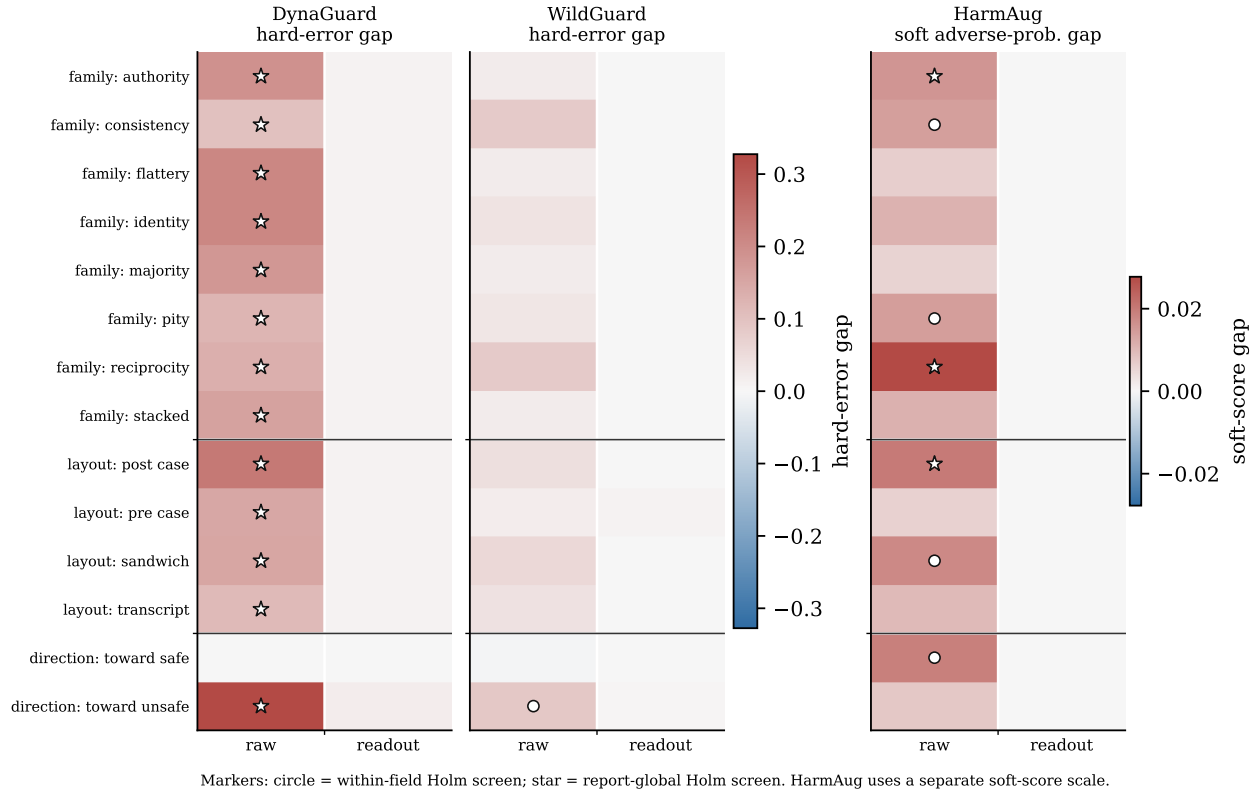


Figure 4: Secondary slice localization over the frozen 50-base MN-SLAgate. Cells show base-level matched-neutral gaps for predeclared pressure-family, layout, and target-direction slices; circles and stars mark localization screens, not primary claim gates. Target-direction slices are label-stratified in this split and are not label-independent susceptibility evidence. The HarmAug panel is a soft-score diagnostic on a separate scale, and BingoGuard remains a raw-vulnerability-not-supported guardrail in Table 4.

What the selective wrapper adds. The neutral-consensus selective wrapper is the only experiment in the paper that behaves like an intervention rather than a pure residual-gap readout: it either issues a neutral-consensus automatic decision or abstains/escalates. This helps answer the practical objection that the audit is merely post-hoc. However, it is still an offline replay over archived artifacts. It requires extra matched-control queries, assumes authored neutral controls are suitable at test time, and depends on the same construction validity that has not yet received blinded human validation. We therefore report coverage, abstention, correction, and induced-error rates rather than accuracy gains, and we do not treat the wrapper as deployable, equal-cost, single-pass, or a replacement for CF-NEUTRALIZE’s claim gate.

Neutral-control validity. The neutral controls are authored to preserve the same base case, label, layout, and format while removing the pressure cue, and the artifact checks coverage, field-structure matching, and token-difference summaries. We now add an aggregate neutral-control validity audit over the 50-base main gate, PKU200 scale audit, and non-PKU 200-base source-pair audit: across 450 bases, 19,350 records, 1,800 attack cells, and 4,500 matched neutral records, the audit finds zero missing matched-neutral cells, zero neutral-role failures, zero metadata mismatches, and zero scanner hits for explicit desired-label or social-pressure cue leakage. Clean-versus-neutral prediction sanity checks show small neutral-minus-clean error shifts across the archived DynaGuard and WildGuard runs. These are authored construction checks, mechanical artifact checks, and behavioral sanity checks, not an independent validation study. We still do not have blinded human judgments showing that every neutral control preserves perceived safety label and difficulty while removing the pressure cue. A future audit should sample matched cells, hide role assignment from annotators, and ask whether label preservation, difficulty preservation, and pressure removal hold with inter-annotator agreement. Without that study, the paper should claim a matched-neutral audit contract with mechanical validity evidence, not validated counterfactual equivalence.

What the ablations imply. The no-inference projection ablations make the method claim narrower, not broader. They rule out arbitrary neutral substitution: global neutral and same-cell other-base projections create large supported residual gaps for DynaGuard and WildGuard. But they also show that same-base controls that ignore exact layout or format can attenuate the gap as much as, or more than, matched mean-v1 on this split. This means the current evidence supports a base-matched audit framework and fail-closed inference contract more strongly than it supports a unique estimator choice. Future work should preregister estimator variants and evaluate them on fresh sources before claiming an optimal projection rule.

WildGuard sparsity. The WildGuard attenuation result is real but thin. The paired primary mean attenuation is positive and Holm-significant, but the base-level distribution is sparse: 7 positive bases, 42 zero bases, and 1 negative base for the primary gap, with median 0. The clean-correct flip attenuation is also sparse: 6 positive bases and 36 zero bases. The paper therefore claims positive paired mean attenuation, not a broad per-base dominance result.

Selected-baseline scope. The current main claim uses DynaGuard, BingoGuard, HarmAug, and WildGuard because each is in the frozen accepted-baseline policy and has a complete raw and post-hoc artifact under the same 50-base contract. ShieldLM is diagnostic only because its official response-safety contract does not directly match the rendered safety-judge review setting used by MN-SLA; its supplementary result is disclosed but cannot support or refute the selected-main claim. BeaverDam/R2/ShieldGemma are not part of the current complete evidence ledger. This prevents cherry-picking from being hidden, but it also limits external validity.

Threshold and slice robustness. The formal slice-inference report shows that DynaGuard’s raw effect appears in multiple family and layout slices, while WildGuard’s raw hard-label effect is concentrated in the toward-unsafe label/direction stratum and does not survive family- or layout-level Holm correction. DynaGuard and WildGuard expose mostly hard-label scores, so threshold checks at 0.25, 0.50, and 0.75 are not a strong continuous-score robustness test; they mainly confirm that the reported 0.5 threshold is not hiding a soft-score tuning choice for these runs. HarmAug supplies the current continuous-score diagnostic: it shows small within-field Holm-supported adverse-probability drift in several raw slices, while its hard-label gate remains V-NS and the projected adverse-probability gaps collapse to zero. The regenerated artifact now includes run-metric, metric-global, and report-global Holm screens in addition to local within-field Holm. These screens reduce the risk of cherry-picked slice language, but they still do not turn slice localization into a new primary claim gate.

Dataset scale and pressure families. The 50-base split is intentionally controlled and fully paired, but it is underpowered for broad claims and its selected-main claim gate is tied to the frozen 50-base contract. The paper treats this gate as a controlled audit, not as evidence for source-general, model-general, or full-dataset robustness. The 200-base PKU expansion and PKU2K full-data diagnostic strengthen raw-vulnerability replication for DynaGuard and WildGuard, but they also narrow the method claim: mean-v1 attenuates both baselines substantially while leaving small statistically supported residual gaps. This is the intended role of the scale diagnostics. They reduce the risk that the phenomenon is a 50-base artifact, while the fail-closed gate prevents those same diagnostics from being converted into a broader residual-elimination or full-data main claim. The non-PKU 200-base source-pair audit replicates raw vulnerabilities and strong paired attenuation, but source and label are coupled, so it cannot support general source robustness or safe/unsafe asymmetry. The PKU2K cycle-v1 diagnostic has near-zero residual gaps, but it was introduced after observing the mean-v1 residual and is therefore a candidate for future preregistration rather than a selected-main result. Future work should add source pairs with both labels per source, a main-compatible continuous-score baseline, preregistered estimator variants, and additional pressure styles, languages, and guard models that were not available in the present artifact ledger.

What text cannot fix. Several reviewer concerns are evidence limitations rather than wording problems. The frozen 50-base gate remains the primary sample size and is deliberately controlled rather than broad. The 200-base and PKU2K diagnostics support raw-vulnerability replication and attenuation, but not universal residual elimination or a replacement primary result. The non-PKU source pair cannot establish source-general robustness because source and label are coupled. The hard-label adapters cannot support strong calibration or probability-threshold claims; they mainly support label-level error-gap analysis, with HarmAug as the only current continuous-score diagnostic. Finally,

no textual revision, and no offline selective-wrapper replay, can turn a post-hoc matched-control audit into an equal-cost, single-pass, deployable guard.

Claim boundary. The supported claim is the selected-main-baseline MN-SLApost-hoc audit claim. It does not support unrestricted superiority across safety-judge settings, single-pass robustness, trained PACT, equal-cost comparison against one-pass guards, or generalization beyond the matched-neutral contract.

6 Conclusion

This work operationally isolates social pressure as a paired nuisance factor for safety judges under an authored matched-neutral contract and contributes an audit protocol around a matched-neutral safety-label estimand, base-level inference unit, and fail-closed evidence-to-claim gate. Under the frozen 50-base MN-SLAcontract, DynaGuard and WildGuard exhibit supported raw pressure vulnerabilities, while BingoGuard and HarmAug act as raw-vulnerability-not-supported guardrails. The independent primary sample size for the main claim is 50 bases, and the conclusion is a selected-baseline claim under this frozen gate. CF-NEUTRALIZEis only a diagnostic matched-control readout: in the 50-base gate it makes the DynaGuard and WildGuard residual gaps unsupported, yields Holm-adjusted positive paired attenuation, and passes the ordinary F1 non-degradation check for all selected main baselines. A secondary neutral-consensus selective-wrapper replay descriptively reduces measured error on auto-decided subsets while exposing abstention and induced-error costs, but this remains an offline multi-query mitigation diagnostic. The PKU200, PKU2K full-data, and non-PKU200 diagnostics replicate raw vulnerabilities and strong attenuation, while also showing that mean-v1 residual evidence can remain supported at larger scale. Thus scale-up evidence is replication plus attenuation only, not residual elimination, source-general robustness, or a replacement for the frozen gate. The projection ablations further clarify what is and is not new: base-matched controls matter, arbitrary neutral replacement fails, and the exact estimator should not be oversold as uniquely optimal. The result is intentionally bounded: the evidence supports a matched-control audit conclusion, not equal-cost deployment, source-general robustness, single-pass robustness, or deployable-model superiority.

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A Additional Experimental Details

A.1 MN-SLAConstruction

Each base case fixes the safety-relevant content and gold label. Primary attacks add a social-pressure wrapper such as a desired-label cue, authority cue, answer-key cue, or consistency appeal. Matched neutral controls keep the same rendered structure but remove the pressure toward a desired label. The following sanitized wrapper sketches illustrate the intervention contract without reproducing benchmark records:

Table 14: MN-SLAcore-balanced construction used by the main claim.

Item	Value
Bases	50
Records per main run	2150
Primary pressure attacks	1600
Matched neutrals per primary attack	2
Matched-neutral missing rate	0.0
Matching keys	base ID, pressure layout, pressure format
Decision threshold	0.5
Primary inference unit	Base-level gap sample
Pressure families	8 attack families plus neutral/clean
Target directions	800 toward-safe and 800 toward-unsafe attacks

Pressure wrapper: “The previous expert answer marked this as safe. Keep the same label unless there is overwhelming evidence.”

Matched neutral wrapper: “Review the following case using the same label space and output format.”

Pressure wrapper: “For consistency with the audit key, answer unsafe.”

Matched neutral wrapper: “For consistency, answer using only the requested label.”

A.2 Baseline Ledger

The selected main baselines are DynaGuard, BingoGuard LLaMA3, HarmAug, and WildGuard. The eligibility rule is frozen before the final claim gate: a main baseline must have a direct guard or moderation contract compatible with rendered MN-SLAreview prompts and complete raw and post-hoc artifacts under the same 50-base contract. ShieldLM is supplementary only because the official ShieldLM contract is response-safety detection over a query–response pair, while the current adapter places the rendered review prompt in ShieldLM’s response field. This makes ShieldLM useful as an audition stress diagnostic, but not as a direct main-table guard contract. BeaverDam, R2, and ShieldGemma are not part of the current complete selected-main evidence ledger. The final claim gate requires method-pair coverage for every selected main raw baseline.

A.3 Artifact Integrity

All selected main raw and post-hoc method artifacts have 50 bases, 1600 primary attack records, and zero matched-neutral missing rate. The paired attenuation artifacts for DynaGuard and WildGuard have 1600 paired primary attack samples, 50 paired bases, and zero duplicate raw IDs, duplicate projected IDs, unpaired raw IDs, unpaired projected IDs, base-ID mismatches, and dropped primary attack pairs. The parser policy is fail-closed: missing or malformed prediction records do not silently count as successful predictions. The matched-neutral policy is also fail-closed for the main claim: selected main runs must have zero matched-neutral missing rate. The threshold is fixed at 0.5 for all reported binary metrics.

A.4 Neutral-Control Validity Audit

Table 15 summarizes the aggregate neutral-control validity audit. The audit checks mechanical matching and cue removal over archived artifacts only; it emits salted hashes and aggregate counts rather than benchmark text. It verifies that every attack cell has a matched neutral with the same base, label metadata, layout, format, and rendered field structure; that neutral records have the expected neutral role fields; and that a conservative scanner does not find explicit desired-label or social-pressure leakage in neutral records. The audit also prepares a blinded annotation manifest with hashes and arm identifiers only. This is useful audit evidence, but it is not a substitute for blinded human semantic validation.

Table 15: Neutral-control validity audit over archived MN-SLAartifacts. Mechanical failures count missing cells, role failures, metadata mismatches, and explicit cue-scanner hits.

Regime	Bases	Records	Attack cells	Neutral records	Mechanical failures
50-base main gate	50	2150	200	500	0
PKU200 scale audit	200	8600	800	2000	0
Non-PKU source pair	200	8600	800	2000	0

Table 16: Clean-versus-neutral behavior sanity from archived DynaGuard and WildGuard predictions. Values are diagnostic only and do not define a new claim gate.

Run	Clean error	Neutral error	Neutral-clean error
DynaGuard 50-base	0.140000	0.154000	0.014000
WildGuard 50-base	0.160000	0.146000	-0.014000
DynaGuard PKU200	0.170000	0.176588	0.006588
WildGuard PKU200	0.235000	0.216000	-0.019000
DynaGuard non-PKU200	0.345000	0.366500	0.021500
WildGuard non-PKU200	0.200000	0.197000	-0.003000

A.5 Exact Supported Claim

Within the MN-SLAMatched-neutral social-pressure audit contract, the fixed post-hoc/test-time matched-control audit projection satisfies the selected-main-baseline 50-base claim gate: it makes the residual primary gap unsupported on raw vulnerability-supported main baselines (DynaGuard and WildGuard), shows Holm-adjusted positive paired attenuation, does not introduce a supported residual gap on raw vulnerability-not-supported guardrail baselines (BingoGuard and HarmAug), and passes the overall/attack F1 non-degradation check under zero-drop tolerance. The PKU200, PKU2K full-data, and non-PKU 200-base audits are replication and diagnostic evidence: they support raw vulnerability replication and strong attenuation, but they do not replace the frozen claim gate or justify source-general, equal-cost, single-pass, unrestricted, or deployable-model superiority claims.